Challenge of cloud platform technology
- changing communication style -

Ikuo Nakagawa

Oct., 2011

EXAGE

INTEC

Vision

Our goal
is a cloud platform

Basic architecture of cloud platform



applications

Dripcast,
open in communities

Developers Env. &
Basic features
(Eclipse or tools)

Dripcast
easy development
framework for Java

Cloud style application component

Mail, SMS & Chat | Voice & Video | Workflow | Document & Calc. | . . . . . .

3rd party modules & libraries

Add-on modules | Add-on modules

Cloud platform API

File System & File Share | Scheduler | Data Processing | Auth / Bill Sessions | . . . . . .

EXAGE by Intec, Inc.

Logger
logging
& analysis

Operation and Management

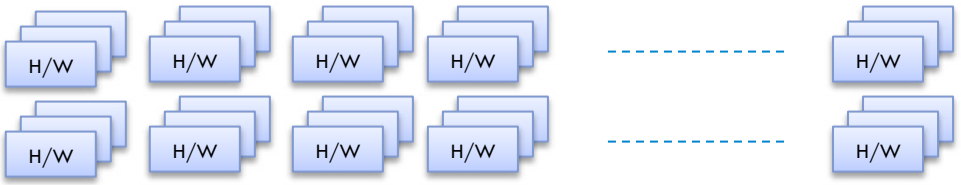
EXAGE / Storage
scale-out, unlimited unified storage
high performance random write.

EXAGE / KVS
scale-out KVS (aka NOSQL)
with very easy Java Map i/f.

EXAGE / Filter
real time / high performance
streaming processing filters

plan
in a future

available
now or soon



H/W, e.g.,
a huge number
of computers

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Technologies

Brief introduction of EXAGE

EXAGE / Storage – Unlimited scale-out storage

2nd generation

Scale-out style storage based on original cloud platform technology by INTEC, with high performance random access I/O (suits for VM or RDB images, as well). It's unified storage, that is, it supports NFS, CIFS and iSCSI.



Features or EXAGE/Storage

1. Unlimited scale-out storage

Just adding storage component (commodity server), to increase storage-space and access performance.

2. High performance random access

100MB/s for GbE, Suitable for VM images for virtualization cloud, or even historical RDB data store.

3. Native file system support

As known as scale-out style "unified storage", which support NFS, CIFS or iSCSI with unlimited file size

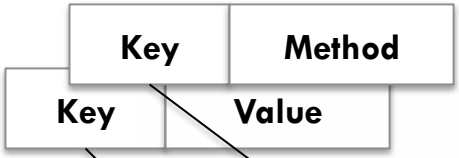
EXAGE / Storage, released in Dec, 2010.
2.2 will be available, soon.

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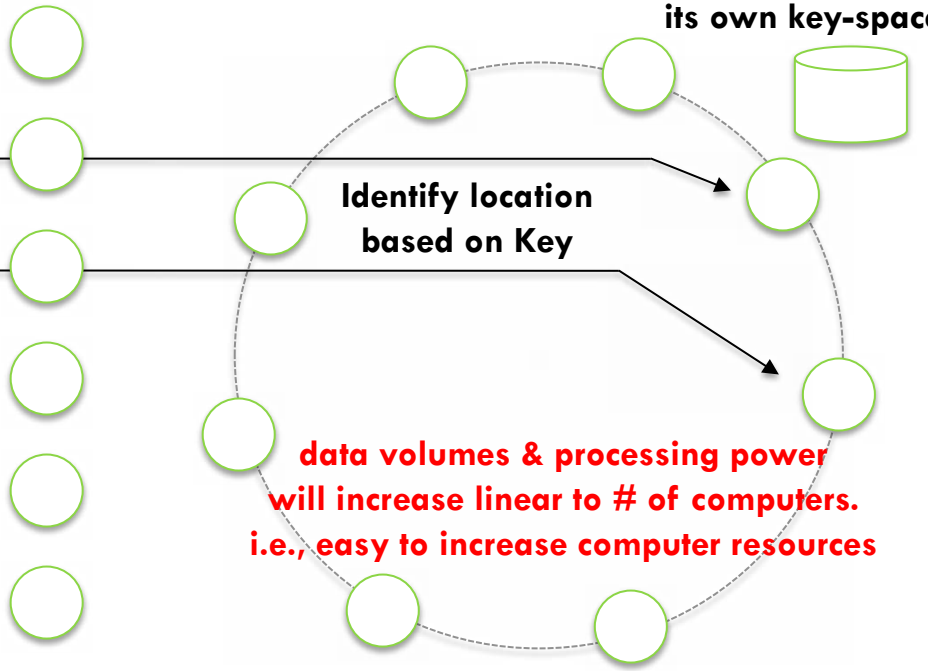
EXAGE / Database – Scale-out, unlimited data processing

Parallel & distributed processing framework, with scale-out style KVS technology.
Dividing "key-spaces", for processing or storing data, among huge # of computers.
Ability to process and to store about 1B objects per node.





not only "storing",
but also "processing", on the cloud.



benefit of scale-out style platform

Easy to plan, design and installation,
just connect a computer to increase power.

Automatic redundancy & failover by S/W,
for scalable and easy operation.

Scale merit by using commodity servers,
for all of computer H/W.

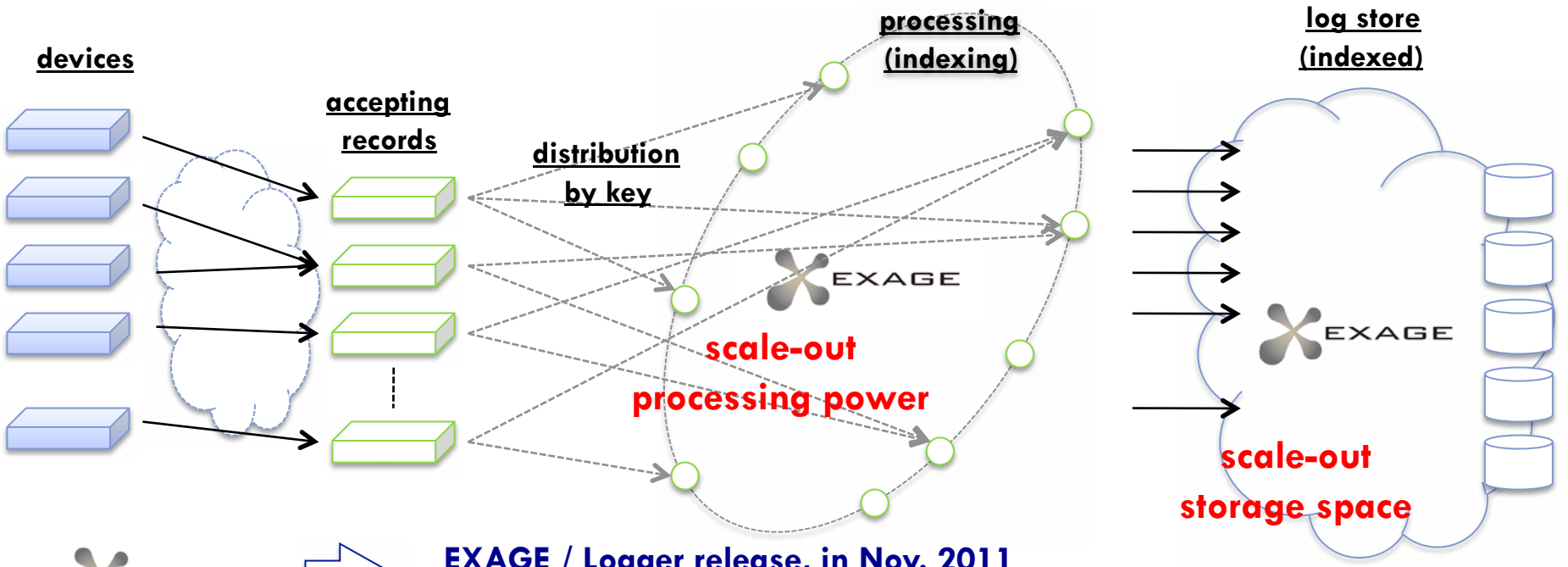
EXAGE / KVS, as known as NoSQL.
will be available, in Nov. 2011

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EXAGE / Filter – Real time stream processing

EXAGE provides scalable processing space for real time stream processing.
It accepts various data or records and process them in parallel & distributed manner.
Logger (with indexing) is one of typical implementation of filters.

data or records (input)	accept & deliver based on keys (distribution)	parallel & distributed real time processing (stream processing)	able to write intoto backend storage (EXAGE / Storage)



EXAGE

EXAGE / Logger release, in Nov. 2011
EXAGE / Filter will be available, 1Q of 2012

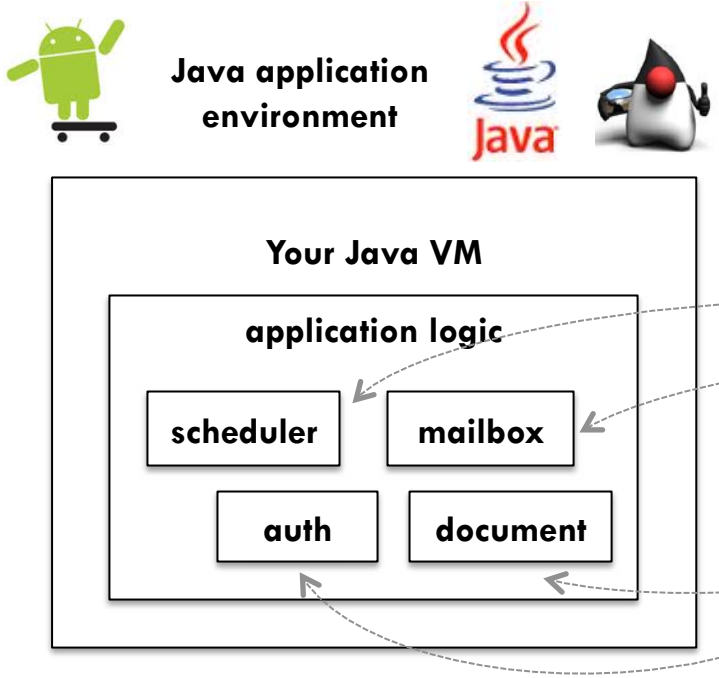
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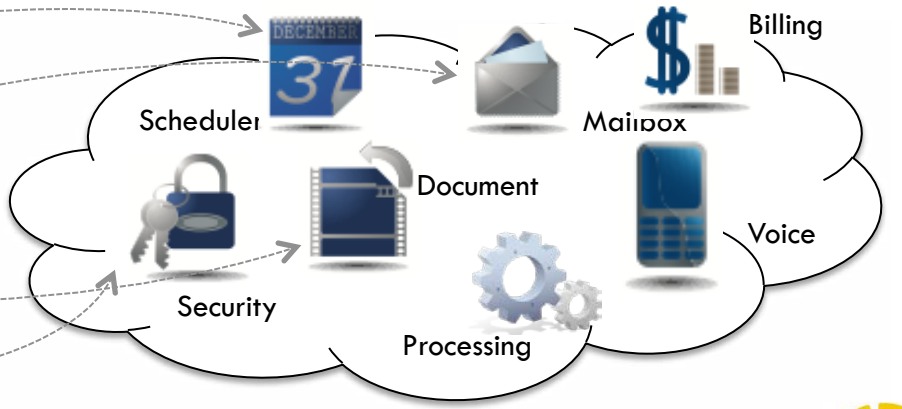
Dripcast – Simple and easy Java framework

Seamless access to data objects on the cloud (almost same as a object, in local)
No need to think about location of the data, no need to write communication (nor HTTP),
Just call Java method simply, to access or process your data object on the cloud.

cutting-edge model, incredibly easy than, Google, Apple or MS

very simple and easy application framework, enables application development to be simple, efficient and productive.





Experimental research project, started with University of Tokyo.
with academic organizations, communities, service operators and application vendors.

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Dripcast

10 times easier development way,
for your "Android+cloud" applications.

Magic "spell" changes the world

"Bibbidi-Bobbidi-Boo"

(also called "The Magic Song")

is a novelty song, written in 1948
by Al Hoffman, Mack David, and Jerry Livingston.

It was introduced in the 1950 Disney film ***Cinderella***,
performed by actress Verna Felton.

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Bibbidi-Bobbidi-Boo

write your code via interface. it's Java !

Try, 3 minutes hacking...

defining "how to access" your data object.

```
// an instance we want to handle.
Gadget a = new GadgetImpl();



// store out value to the object, at first.
a.setValue(31);

// now, verify input against our value.
String s = in.readLine();
if (Integer.parseInt(s) == a.getValue()) {
  System.out.println("Great! Lucky guess!");
} else {
  System.out.println("No match. Sorry.");
}
```

```
interface Gadget {
  void setValue(int val);
  int getValue();
}
```

```
class GadgetImpl implements Gadget {
  private int value = 0;
  public void setValue(int val) {
    value = val;
  }
  public int getValue() {
    return value;
  }
}
```

defining "what to do" when you call it.

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Bibbidi-Bobbidi-Boo

write your code via interface. it's Java !

Try, 3 minutes hacking...

defining "how to access" your data object.

```
// an instance we want to handle.
Gadget a = new GadgetImpl();

// Biddidi-Bobbidi-Boo.
a = B3.setAndAttachInstance(
        key, Gadget.class, a);

// store out value to the object, at first.
a.setValue(31);

// now, verify input against our value.
String s = in.readLine();
if (Integer.parseInt(s) == a.getValue()) {
  System.out.println("Great! Lucky guess!");
} else {
  System.out.println("No match. Sorry.");
}
```

creating your data object on a cloud.

executing your method (request) on a cloud, as well.

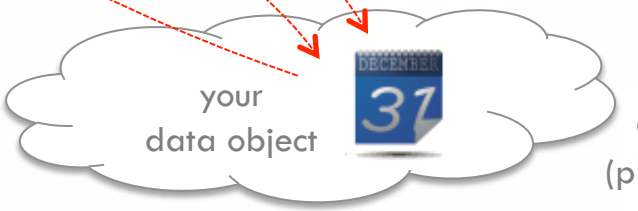
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interface Gadget {
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class GadgetImpl implements Gadget {
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    return value;
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}
```

defining "what to do" when you call it.



Android device (application)



Cloud (platform)

What you need to do, is only casting a spell "Bibbidi-Babbidi-Boo" !

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Try it !

So, what's happen in Android application world ?
Does it make any changes ?

Example

Develop a group communication tool for Android.

for example,

Let's share GPS location information,
between group members.

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Example of "sharing GPS location" application

It takes only 3 minutes to implement a stoker application to share GPS info via the cloud.
You only need to define class to identify a location (in Java), and share an instance.
No table definition nor REST/HTTP implementation would be required.

```
public class YourActivity implements Activity, LocationListener {
  UUID key = UUID.fromString("your-group-UUID"); // to share an object between group members.
  NavigableMap map; // will be associated with a cloud object.
  String id; // to store my phone number.

  public void onCreate() {
    map = B3.createOrAttachInstance(key, NavigableMap.class, "java.util.TreeMap");
    id = ((TelephonyManager)getSystemService(TELEPHONY_SERVICE)).getLine1Number();
    ((LocationManager)getSystemService(LOCATION_SERVICE)).requestLocationUpdates(
      LocationManager.GPS_PROVIDER, 0, 0, this);
  }

  public void onLocationChanged(Location location) {
    int x = (int)(location.getLatitude() * 1E6);
    int y = (int)(location.getLongitue() * 1E6)
    map.put(id, x + "," + y);

    // show, your members positions....
    Entry entry = map.firstEntry();
    while (entry != null) {
      // show id and location : entry.getValue()
      entry = map.higherEntry(entry.getKey());
    }
  }
}
```



unique object
on a cloud

GPS info

your-group-UUID

store your GPS info

getting GPS info
of your party members

our models are ;

cloud-full computing

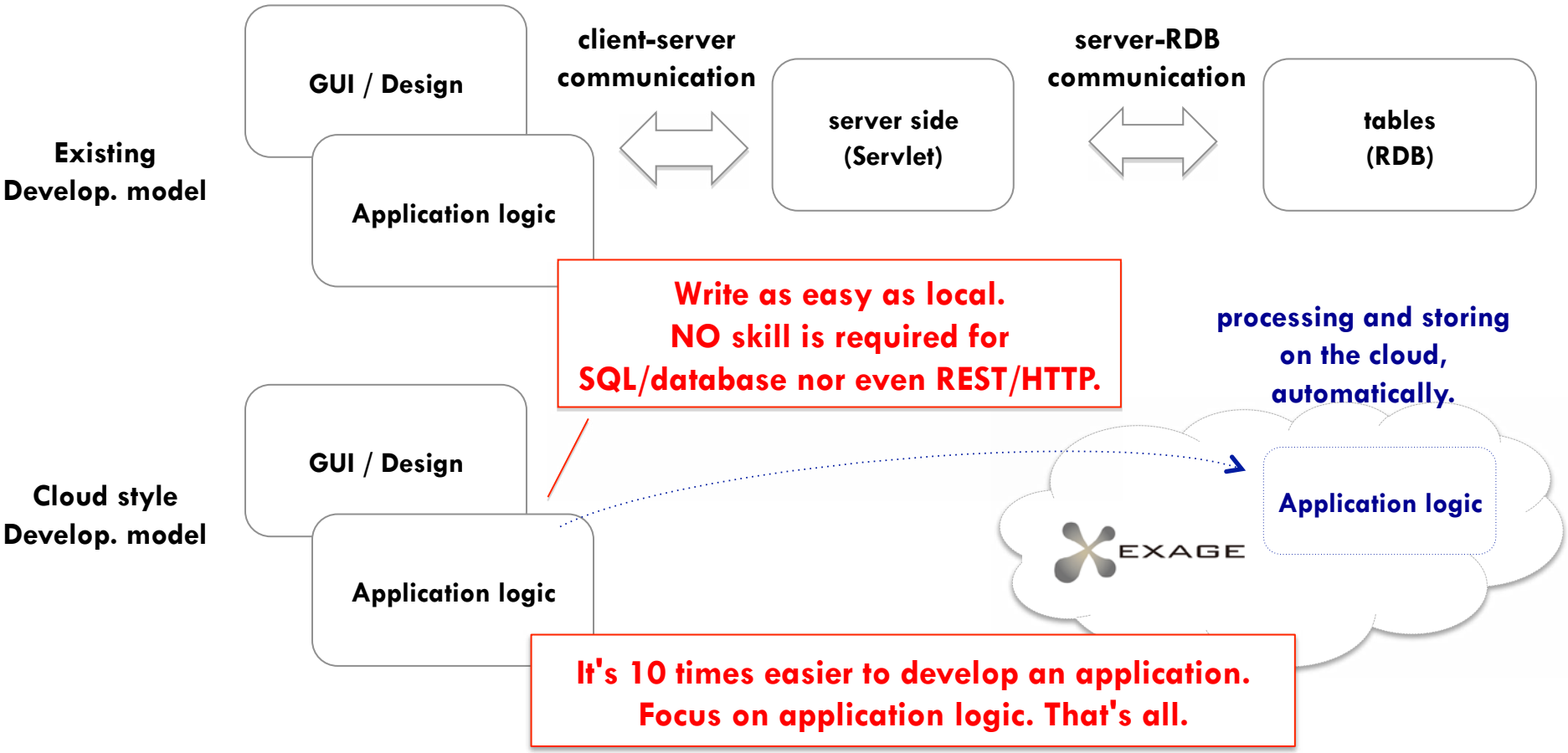
and

server-less programming

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Server-less programming – forget about server side

There is NO need to develop server side processing nor database programming.
Our "Spell" delivers Java object into the cloud, both of for storing and processing.
You need only to develop application logic. It's very simple and efficient.



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No more servers nor back-end systems for developers

Of course, there is no need to use SQL/RDB, Servlet nor REST/HTTP.
that means, we don't have to learn more about,...
Oracle, MySQL, PostgreSQL, Ruby, python, php, apache or others.

No more server side technologies



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Considerations about networking

Challenge

Forget about servers !

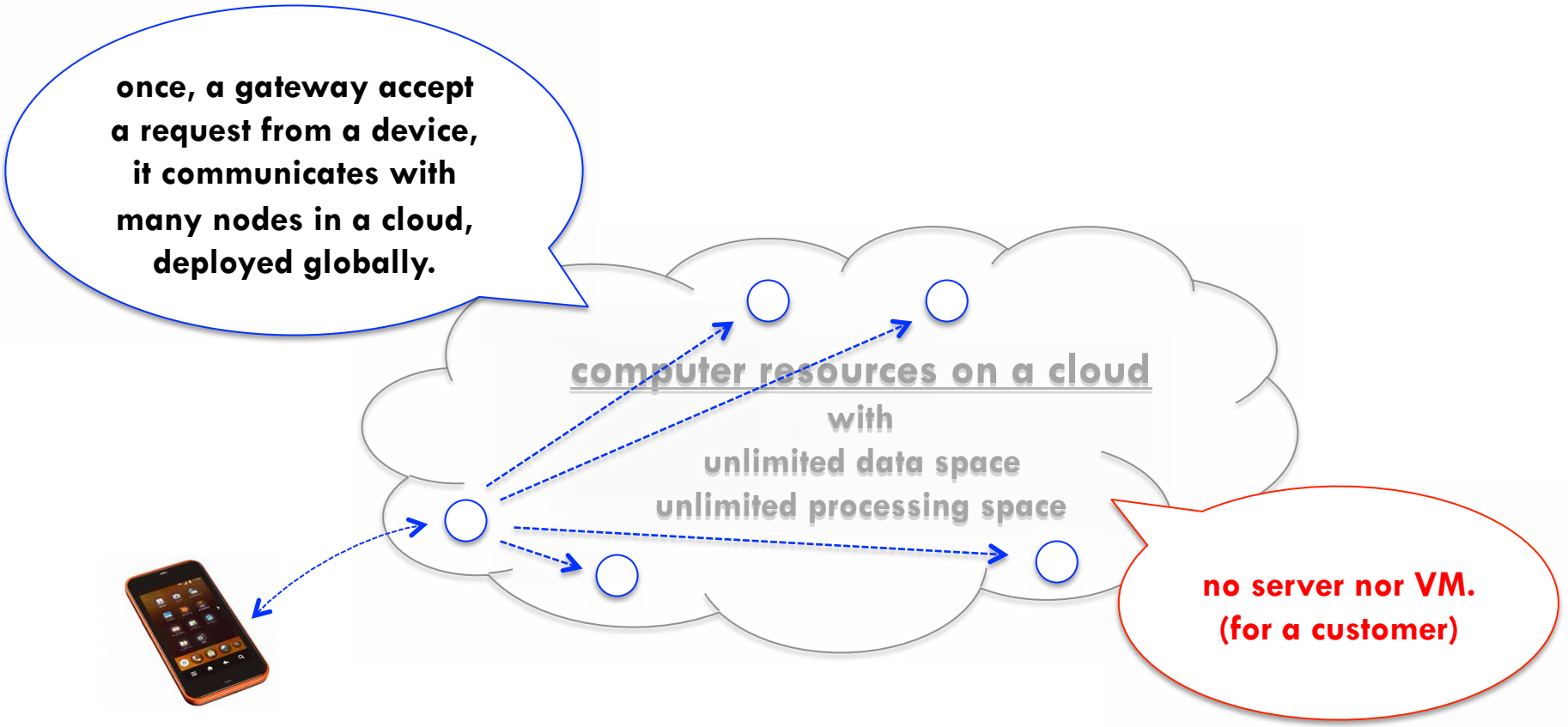
No server nor virtual machine for customers.

Customers know only about platform.

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Communication in the new computing model

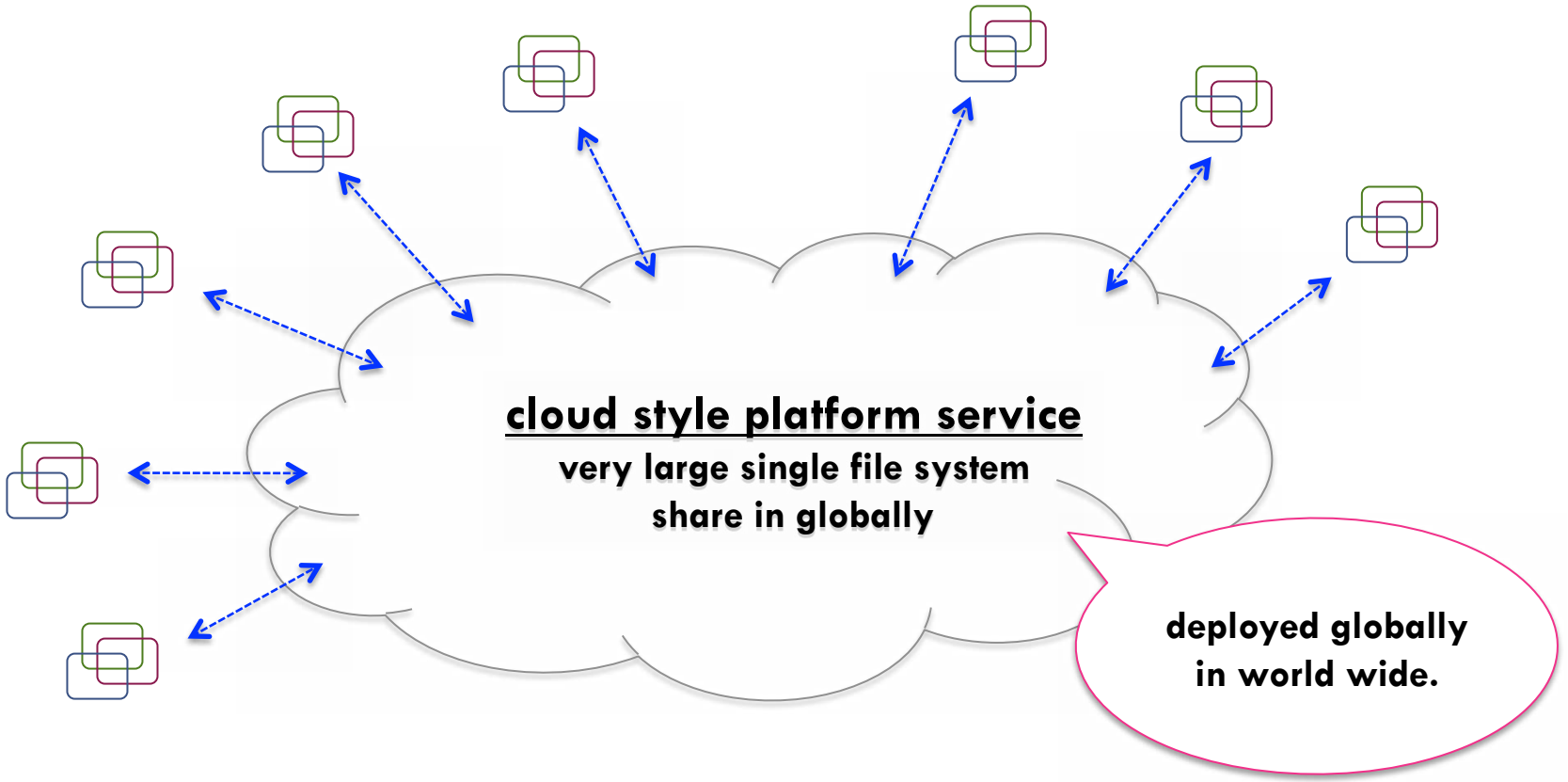
Device + cloud computing model, also changes global communication model.
A device send a request to a nearest gateway of a cloud, and
the gateway communicates with many nodes in a cloud, deployed globally.



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A simple example – Use case of platform technology

Think about cloud style platform service, as a simple example,
Customers can store and process your data object on a cloud from any location of the world.
(cloud platform has a very large single data space, shared in globally)

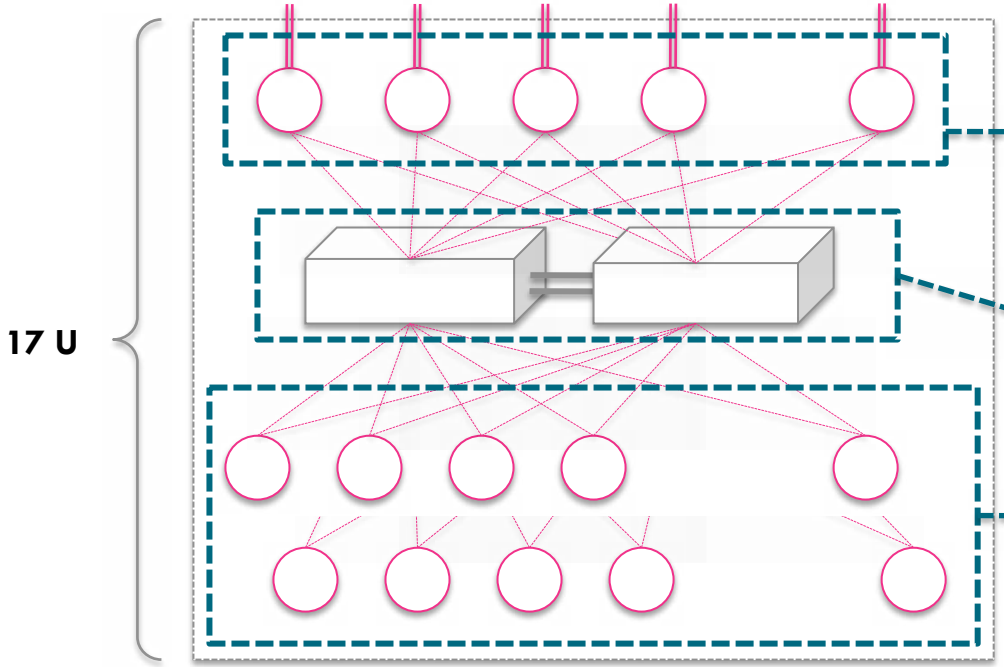


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Typical configuration (a unit)

In a typical configuration, we install 10 access servers and 25 core servers in an "unit", using distributed LAG (link aggregation group) with very simple L2 configuration. We provides about 10Gbps for external, with 30Gbps internal traffic.

typical configuration (an unit)



expecting throughput

external service network

up to 10Gbps for external

10 Access Server with bonding

av. 10Gbps for internal

36 port L2-SW x 2 with DLAG

up to 30Gbps for internal

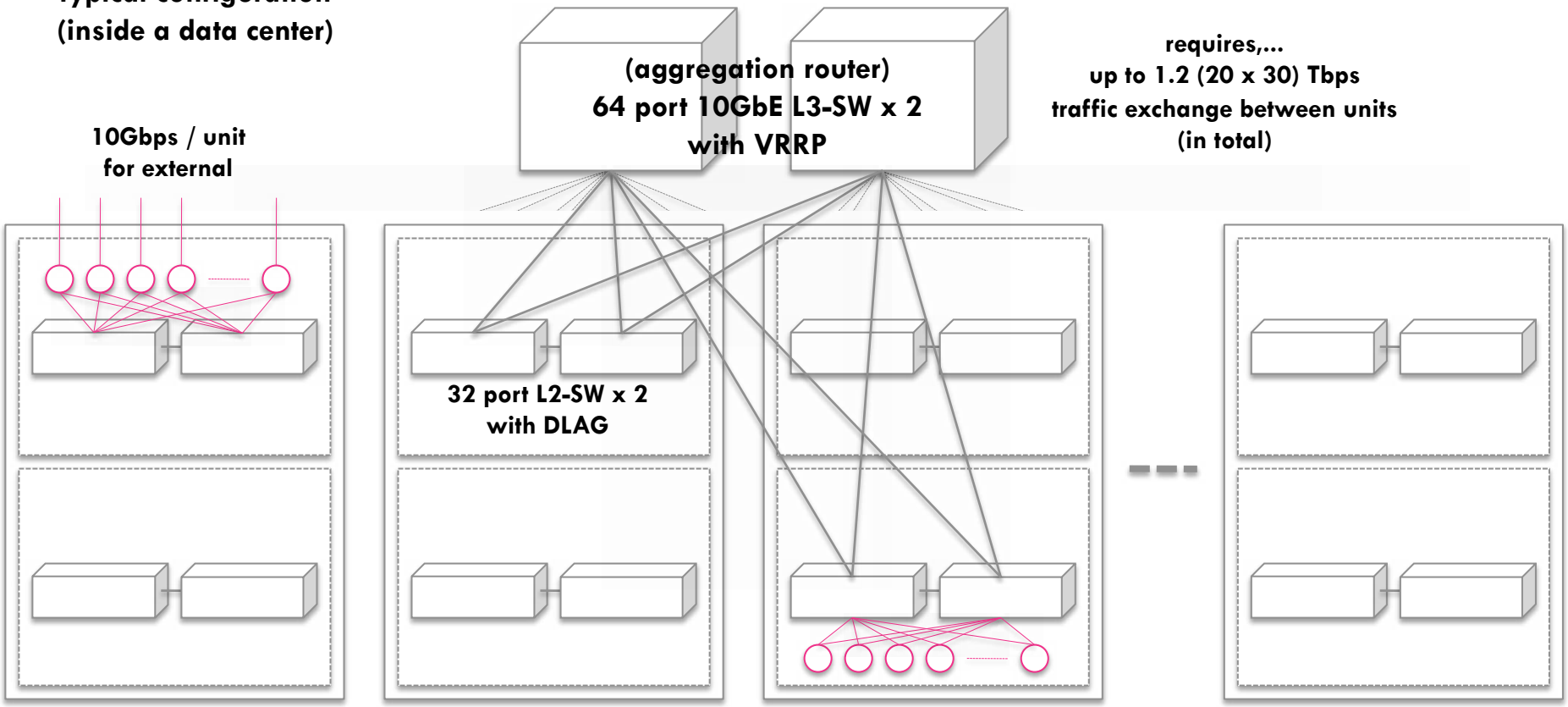
25 Core Server with bonding

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Typical configuration (data center)

Nowadays, customers want about 1PB data space, which requires 750 core servers.
in our configuration, it requires 30 units (15 racks) in a data center.
we need very high speed aggregation router (L3-SW) for 600 Gbps internal traffic.

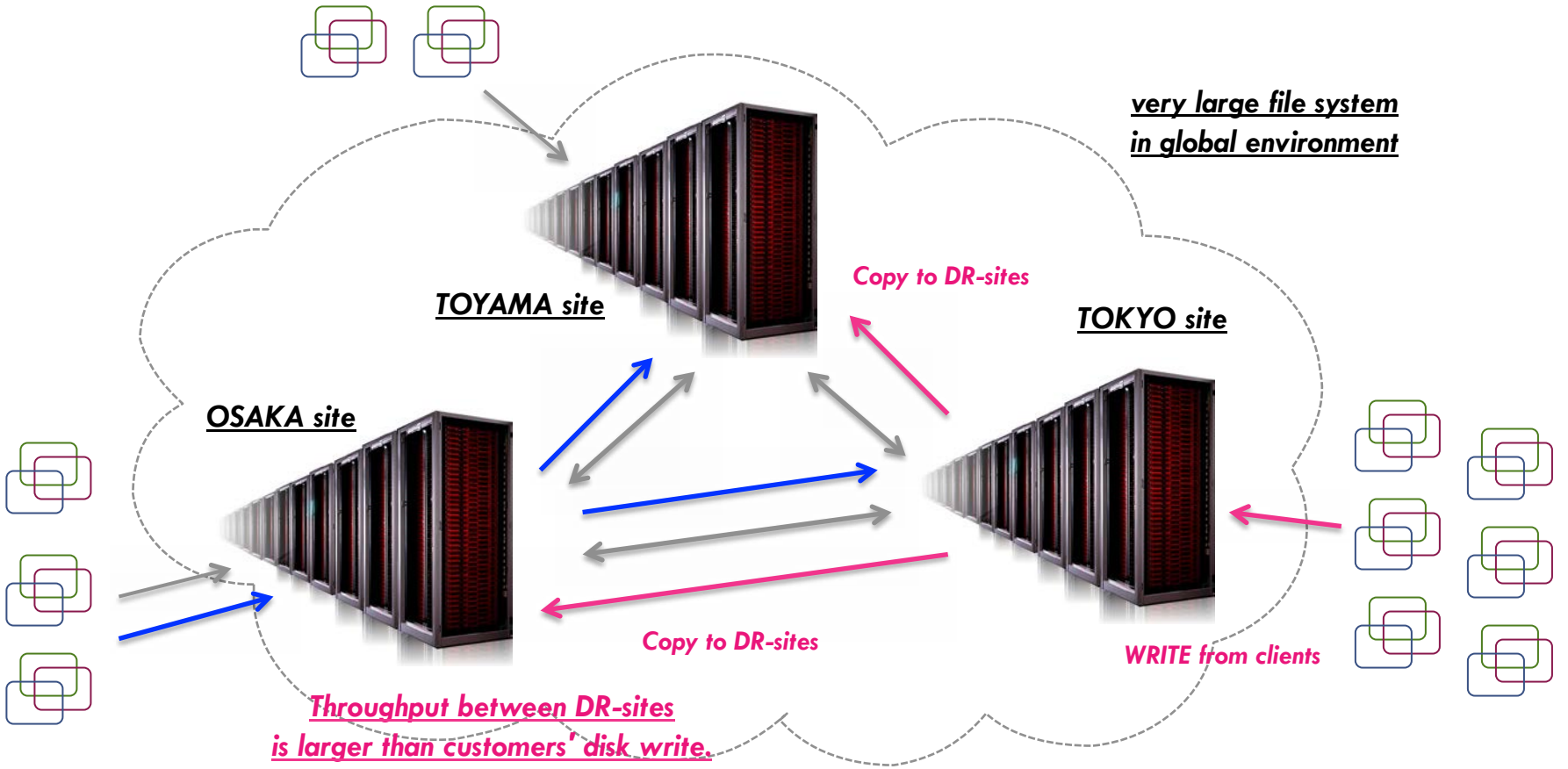
typical configuration
(inside a data center)



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More configuration (between DR sites)

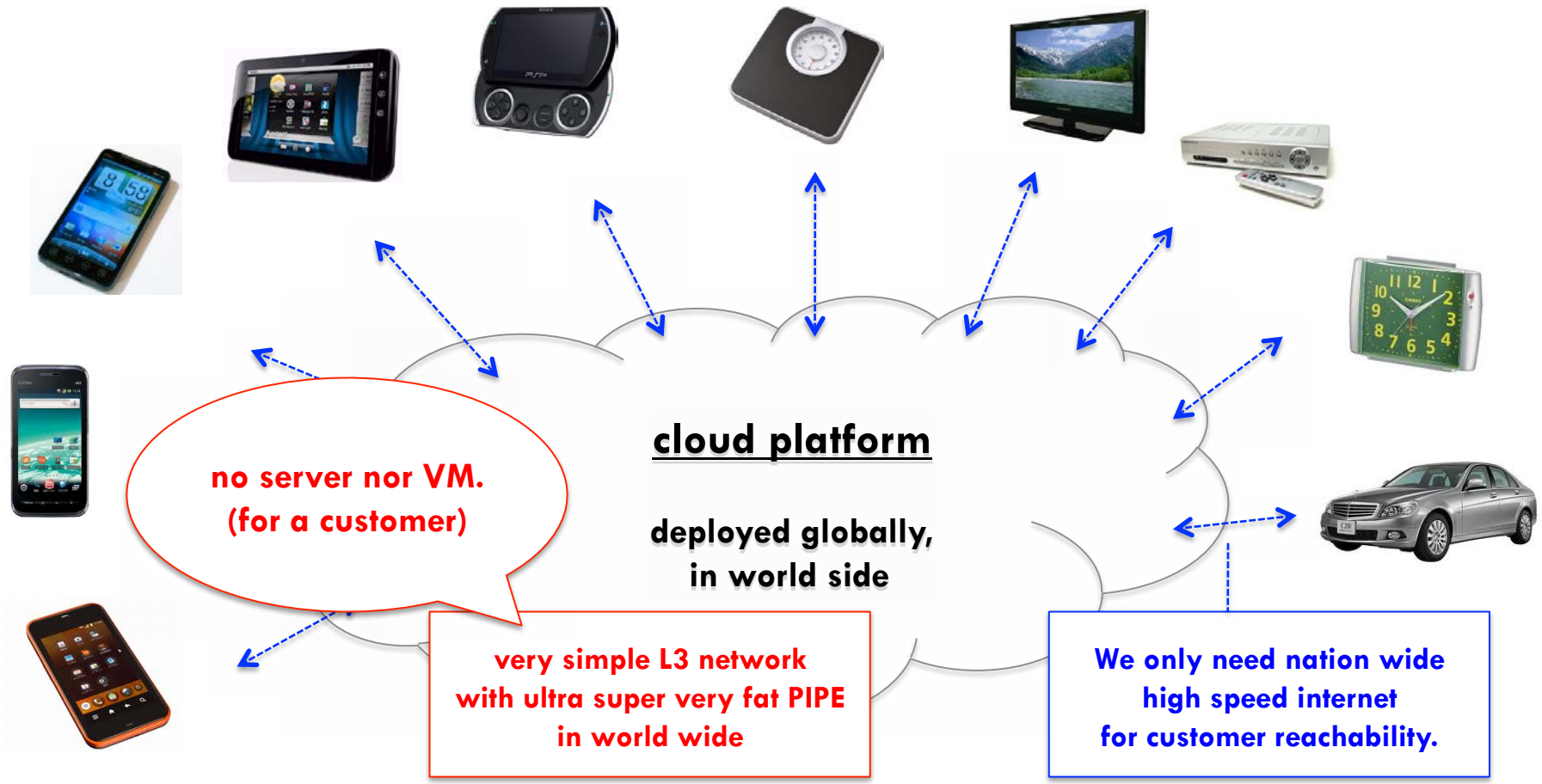
DR (disaster recovery) is most prioritized requirement of cutting-edge cloud service operators.
In technically, we can do that. But we need ultra super very fat pipe between service sites.
For real time DR mechanism, we need more throughput between site, of course.



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Considerations

We need high speed nation wide internet for smartphone or device nodes.
In addition, we need ultra super very fat PIPE for our internal networks for cloud platform, where cloud platform would be deployed globally, in world wide.



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F.Y.I.

F.Y.I. - Research project started for Dripcast

Research project for server-less programming model



"live on cloud, play on device"

Action items :

1. Research a new "server-less" programming model.
optimize data model & processing model of direct access to cloud objects.

2. Deploy and standardize very easy and simple development model.
For several languages and environments. "dripcast" is one of such impl. for Java.

3. Evaluate and feedback of new style cloud platform service.
Application (providers, developers) and Platform (network, IDC, cloud operators).

Demonstration site is available.
Visit http://dripcast.org
for more details.

Thanks for supporting our project.
Smart Technologies, A.T.Works,
NOAH Cloud (IDCF),
and many organizations...



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